

**A First Course in
Design and Analysis
of Experiments**

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Chapter 1

Introduction

Researchers use experiments to answer questions. Typical questions might be:

Experiments
answer questions

- Is a drug a safe, effective cure for a disease? This could be a test of how AZT affects the progress of AIDS.
- Which combination of protein and carbohydrate sources provides the best nutrition for growing lambs?
- How will long-distance telephone usage change if our company offers a different rate structure to our customers?
- Will an ice cream manufactured with a new kind of stabilizer be as palatable as our current ice cream?
- Does short-term incarceration of spouse abusers deter future assaults?
- Under what conditions should I operate my chemical refinery, given this month's grade of raw material?

This book is meant to help decision makers and researchers design good experiments, analyze them properly, and answer their questions.

1.1 Why Experiment?

Consider the spousal assault example mentioned above. Justice officials need to know how they can reduce or delay the recurrence of spousal assault. They are investigating three different actions in response to spousal assaults. The

Treatments,
experimental
units, and
responses

assailant could be warned, sent to counseling but not booked on charges, or arrested for assault. Which of these actions works best? How can they compare the effects of the three actions?

This book deals with *comparative experiments*. We wish to compare some *treatments*. For the spousal assault example, the treatments are the three actions by the police. We compare treatments by using them and comparing the outcomes. Specifically, we apply the treatments to *experimental units* and then measure one or more *responses*. In our example, individuals who assault their spouses could be the experimental units, and the response could be the length of time until recurrence of assault. We compare treatments by comparing the responses obtained from the experimental units in the different treatment groups. This could tell us if there are any differences in responses between the treatments, what the estimated sizes of those differences are, which treatment has the greatest estimated delay until recurrence, and so on.

An experiment is characterized by the treatments and experimental units to be used, the way treatments are assigned to units, and the responses that are measured.

Advantages of
experiments

Experiments help us answer questions, but there are also nonexperimental techniques. What is so special about experiments? Consider that:

1. Experiments allow us to set up a direct comparison between the treatments of interest.
2. We can design experiments to minimize any bias in the comparison.
3. We can design experiments so that the error in the comparison is small.
4. Most important, we are in control of experiments, and having that control allows us to make stronger inferences about the nature of differences that we see in the experiment. Specifically, we may make inferences about *causation*.

Control versus
observation

This last point distinguishes an experiment from an *observational study*. An observational study also has treatments, units, and responses. However, in the observational study we merely observe which units are in which treatment groups; we don't get to control that assignment.

Example 1.1

Does spanking hurt?

Let's contrast an experiment with an observational study described in Straus, Sugarman, and Giles-Sims (1997). A large survey of women aged 14 to 21 years was begun in 1979; by 1988 these same women had 1239 children

between the ages of 6 and 9 years. The women and children were interviewed and tested in 1988 and again in 1990. Two of the items measured were the level of antisocial behavior in the children and the frequency of spanking. Results showed that children who were spanked more frequently in 1988 showed larger increases in antisocial behavior in 1990 than those who were spanked less frequently. Does spanking cause antisocial behavior? Perhaps it does, but there are other possible explanations. Perhaps children who were becoming more troublesome in 1988 may have been spanked more frequently, while children who were becoming less troublesome may have been spanked less frequently in 1988.

The drawback of observational studies is that the grouping into “treatments” is not under the control of the experimenter and its mechanism is usually unknown. Thus observed differences in responses between treatment groups could very well be due to these other hidden mechanisms, rather than the treatments themselves.

It is important to say that while experiments have some advantages, observational studies are also useful and can produce important results. For example, studies of smoking and human health are observational, but the link that they have established is one of the most important public health issues today. Similarly, observational studies established an association between heart valve disease and the diet drug fen-phen that led to the withdrawal of the drugs fenfluramine and dexfenfluramine from the market (Connolly et al. 1997 and US FDA 1997).

Observational
studies are useful
too

Mosteller and Tukey (1977) list three concepts associated with causation and state that two or three are needed to support a causal relationship:

Causal
relationships

- Consistency
- Responsiveness
- Mechanism.

Consistency means that, all other things being equal, the relationship between two variables is consistent across populations in direction and maybe in amount. Responsiveness means that we can go into a system, change the causal variable, and watch the response variable change accordingly. Mechanism means that we have a step-by-step mechanism leading from cause to effect.

In an experiment, we are in control, so we can achieve responsiveness. Thus, if we see a consistent difference in observed response between the various treatments, we can infer that the treatments caused the differences in response. We don’t need to know the mechanism—we can demonstrate

Experiments can
demonstrate
consistency and
responsiveness

causation by experiment. (This is not to say that we shouldn't try to learn mechanisms—we should. It's just that we don't need mechanism to infer causation.)

Ethics constrain experimentation

We should note that there are times when experiments are not feasible, even when the knowledge gained would be extremely valuable. For example, we can't perform an experiment proving once and for all that smoking causes cancer in humans. We can observe that smoking is associated with cancer in humans; we have mechanisms for this and can thus infer causation. But we cannot demonstrate responsiveness, since that would involve making some people smoke, and making others not smoke. It is simply unethical.

1.2 Components of an Experiment

An experiment has treatments, experimental units, responses, and a method to assign treatments to units.

Treatments, units, and assignment method specify the *experimental design*.

Some authors make a distinction between the selection of treatments to be used, called “treatment design,” and the selection of units and assignment of treatments, called “experiment design.”

Analysis not part of design, but consider it during planning

Note that there is no mention of a method for analyzing the results. Strictly speaking, the analysis is not part of the design, though a wise experimenter will consider the analysis when planning an experiment. Whereas the design determines the proper analysis to a great extent, we will see that two experiments with similar designs may be analyzed differently, and two experiments with different designs may be analyzed similarly. Proper analysis depends on the design and the kinds of statistical model assumptions we believe are correct and are willing to assume.

Not all experimental designs are created equal. A good experimental design must

- Avoid systematic error
- Be precise
- Allow estimation of error
- Have broad validity.

We consider these in turn.

Comparative experiments estimate differences in response between treatments. If our experiment has systematic error, then our comparisons will be biased, no matter how precise our measurements are or how many experimental units we use. For example, if responses for units receiving treatment one are measured with instrument A, and responses for treatment two are measured with instrument B, then we don't know if any observed differences are due to treatment effects or instrument miscalibrations. Randomization, as will be discussed in Chapter 2, is our main tool to combat systematic error.

Design to avoid
systematic error

Even without systematic error, there will be random error in the responses, and this will lead to random error in the treatment comparisons. Experiments are precise when this random error in treatment comparisons is small. Precision depends on the size of the random errors in the responses, the number of units used, and the experimental design used. Several chapters of this book deal with designs to improve precision.

Design to
increase
precision

Experiments must be designed so that we have an estimate of the size of random error. This permits statistical inference: for example, confidence intervals or tests of significance. We cannot do inference without an estimate of error. Sadly, experiments that cannot estimate error continue to be run.

Design to
estimate error

The conclusions we draw from an experiment are applicable to the experimental units we used in the experiment. If the units are actually a statistical sample from some population of units, then the conclusions are also valid for the population. Beyond this, we are extrapolating, and the extrapolation might or might not be successful. For example, suppose we compare two different drugs for treating attention deficit disorder. Our subjects are preadolescent boys from our clinic. We might have a fair case that our results would hold for preadolescent boys elsewhere, but even that might not be true if our clinic's population of subjects is unusual in some way. The results are even less compelling for older boys or for girls. Thus if we wish to have wide validity—for example, broad age range and both genders—then our experimental units should reflect the population about which we wish to draw inference.

Design to widen
validity

We need to realize that some compromise will probably be needed between these goals. For example, broadening the scope of validity by using a variety of experimental units may decrease the precision of the responses.

Compromise
often needed

1.3 Terms and Concepts

Let's define some of the important terms and concepts in design of experiments. We have already seen the terms treatment, experimental unit, and response, but we define them again here for completeness.

Treatments are the different procedures we want to compare. These could be different kinds or amounts of fertilizer in agronomy, different long-distance rate structures in marketing, or different temperatures in a reactor vessel in chemical engineering.

Experimental units are the things to which we apply the treatments. These could be plots of land receiving fertilizer, groups of customers receiving different rate structures, or batches of feedstock processing at different temperatures.

Responses are outcomes that we observe after applying a treatment to an experimental unit. That is, the response is what we measure to judge what happened in the experiment; we often have more than one response. Responses for the above examples might be nitrogen content or biomass of corn plants, profit by customer group, or yield and quality of the product per ton of raw material.

Randomization is the use of a known, understood probabilistic mechanism for the assignment of treatments to units. Other aspects of an experiment can also be randomized: for example, the order in which units are evaluated for their responses.

Experimental Error is the random variation present in all experimental results. Different experimental units will give different responses to the same treatment, and it is often true that applying the same treatment over and over again to the same unit will result in different responses in different trials. Experimental error does not refer to conducting the wrong experiment or dropping test tubes.

Measurement units (or response units) are the actual objects on which the response is measured. These may differ from the experimental units. For example, consider the effect of different fertilizers on the nitrogen content of corn plants. Different field plots are the experimental units, but the measurement units might be a subset of the corn plants on the field plot, or a sample of leaves, stalks, and roots from the field plot.

Blinding occurs when the evaluators of a response do not know which treatment was given to which unit. Blinding helps prevent bias in the evaluation, even unconscious bias from well-intentioned evaluators. Double blinding occurs when both the evaluators of the response and the (human subject) experimental units do not know the assignment of treatments to units. Blinding the subjects can also prevent bias, because subject responses can change when subjects have expectations for certain treatments.

Control has several different uses in design. First, an experiment is *controlled* because we as experimenters assign treatments to experimental units. Otherwise, we would have an observational study.

Second, a *control* treatment is a “standard” treatment that is used as a baseline or basis of comparison for the other treatments. This control treatment might be the treatment in common use, or it might be a null treatment (no treatment at all). For example, a study of new pain killing drugs could use a standard pain killer as a control treatment, or a study on the efficacy of fertilizer could give some fields no fertilizer at all. This would control for average soil fertility or weather conditions.

Placebo is a null treatment that is used when the act of applying a treatment—any treatment—has an effect. Placebos are often used with human subjects, because people often respond to any treatment: for example, reduction in headache pain when given a sugar pill. Blinding is important when placebos are used with human subjects. Placebos are also useful for nonhuman subjects. The apparatus for spraying a field with a pesticide may compact the soil. Thus we drive the apparatus over the field, without actually spraying, as a placebo treatment.

Factors combine to form treatments. For example, the baking treatment for a cake involves a given time at a given temperature. The treatment is the combination of time and temperature, but we can vary the time and temperature separately. Thus we speak of a time factor and a temperature factor. Individual settings for each factor are called *levels* of the factor.

Confounding occurs when the effect of one factor or treatment cannot be distinguished from that of another factor or treatment. The two factors or treatments are said to be confounded. Except in very special circumstances, confounding should be avoided. Consider planting corn variety A in Minnesota and corn variety B in Iowa. In this experiment, we cannot distinguish location effects from variety effects—the variety factor and the location factor are confounded.

1.4 Outline

Here is a road map for this book, so that you can see how it is organized. The remainder of this chapter gives more detail on experimental units and responses. Chapter 2 elaborates on the important concept of randomization. Chapters 3 through 7 introduce the basic experimental design, called

the Completely Randomized Design (CRD), and describe its analysis in considerable detail. Chapters 8 through 10 add factorial treatment structure to the CRD, and Chapters 11 and 12 add random effects to the CRD. The idea is that we learn these different treatment structures and analyses in the simplest design setting, the CRD. These structures and analysis techniques can then be used almost without change in the more complicated designs that follow.

We begin learning new experimental designs in Chapter 13, which introduces complete block designs. Chapter 14 introduces general incomplete blocks, and Chapters 15 and 16 deal with incomplete blocks for treatments with factorial structure. Chapter 17 introduces covariates. Chapters 18 and 19 deal with special treatment structures, including fractional factorials and response surfaces. Finally, Chapter 20 provides a framework for planning an experiment.

1.5 More About Experimental Units

Experimental and
measurement
units

Experimentation is so diverse that there are relatively few general statements that can be made about experimental units. A common source of difficulty is the distinction between experimental units and measurement units. Consider an educational study, where six classrooms of 25 first graders each are assigned at random to two different reading programs, with all the first graders evaluated via a common reading exam at the end of the school year. Are there six experimental units (the classrooms) or 150 (the students)?

Experimental unit
could get any
treatment

One way to determine the experimental unit is via the consideration that an experimental unit should be able to receive any treatment. Thus if students were the experimental units, we could see more than one reading program in each classroom. However, the nature of the experiment makes it clear that all the students in the classroom receive the same program, so the classroom as a whole is the experimental unit. We don't measure how a classroom reads, though; we measure how students read. Thus students are the measurement units for this experiment.

There are many situations where a treatment is applied to group of objects, some of which are later measured for a response. For example,

- Fertilizer is applied to a plot of land containing corn plants, some of which will be harvested and measured. The plot is the experimental unit and the plants are the measurement units.
- Ingots of steel are given different heat treatments, and each ingot is punched in four locations to measure its hardness. Ingots are the experimental units and locations on the ingot are measurement units.

- Mice are caged together, with different cages receiving different nutritional supplements. The cage is the experimental unit, and the mice are the measurement units.

Treating measurement units as experimental usually leads to overoptimistic analysis more—we will reject null hypotheses more often than we should, and our confidence intervals will be too short and will not have their claimed coverage rates. The usual way around this is to determine a single response for each experimental unit. This single response is typically the average or total of the responses for the measurement units within an experimental unit, but the median, maximum, minimum, variance or some other summary statistic could also be appropriate depending on the goals of the experiment.

Use a summary
of the
measurement unit
responses as
experimental unit
response

A second issue with units is determining their “size” or “shape.” For agricultural experiments, a unit is generally a plot of land, so size and shape have an obvious meaning. For an animal feeding study, size could be the number of animals per cage. For an ice cream formulation study, size could be the number of liters in a batch of ice cream. For a computer network configuration study, size could be the length of time the network is observed under load conditions.

Size of units

Not all measurement units in an experimental unit will be equivalent. For the ice cream, samples taken near the edge of a carton (unit) may have more ice crystals than samples taken near the center. Thus it may make sense to plan the units so that the ratio of edge to center is similar to that in the product’s intended packaging. Similarly, in agricultural trials, guard rows are often planted to reduce the effect of being on the edge of a plot. You don’t want to construct plots that are all edge, and thus all guard row. For experiments that occur over time, such as the computer network study, there may be a transient period at the beginning before the network moves to steady state. You don’t want units so small that all you measure is transient.

Edge may be
different than
center

One common situation is that there is a fixed resource available, such as a fixed area, a fixed amount of time, or a fixed number of measurements. This fixed resource needs to be divided into units (and perhaps measurement units). How should the split be made? In general, more experimental units with fewer measurement units per experimental unit works better (see, for example, Fairfield Smith 1938). However, smaller experimental units are inclined to have greater edge effect problems than are larger units, so this recommendation needs to be moderated by consideration of the actual units.

More
experimental
units, fewer
measurement
units usually
better

A third important issue is that the response of a given unit should not depend on or be influenced by the treatments given other units or the responses of other units. This is usually ensured through some kind of separation of the units, either in space or time. For example, a forestry experiment would

Independence of
units

provide separation between units, so that a fast-growing tree does not shade trees in adjacent units and thus make them grow more slowly; and a drug trial giving the same patient different drugs in sequence would include a washout period between treatments, so that a drug would be completely out of a patient's system before the next drug is administered.

When the response of a unit is influenced by the treatment given to other units, we get confounding between the treatments, because we cannot estimate treatment response differences unambiguously. When the response of a unit is influenced by the response of another unit, we get a poor estimate of the precision of our experiment. In particular, we usually overestimate the precision. Failure to achieve this independence can seriously affect the quality of any inferences we might make.

Sample size A final issue with units is determining how many units are required. We consider this in detail in Chapter 7.

1.6 More About Responses

Primary response We have been discussing “the” response, but it is a rare experiment that measures only a single response. Experiments often address several questions, and we may need a different response for each question. Responses such as these are often called *primary* responses, since they measure the quantity of primary interest for a unit.

Surrogate responses We cannot always measure the primary response. For example, a drug trial might be used to find drugs that increase life expectancy after initial heart attack: thus the primary response is years of life after heart attack. This response is not likely to be used, however, because it may be decades before the patients in the study die, and thus decades before the study is completed. For this reason, experimenters use *surrogate* responses. (It isn't only impatience; it becomes more and more difficult to keep in contact with subjects as time goes on.)

Surrogate responses are responses that are supposed to be related to—and predictive for—the primary response. For example, we might measure the fraction of patients still alive after five years, rather than wait for their actual lifespans. Or we might have an instrumental reading of ice crystals in ice cream, rather than use a human panel and get their subjective assessment of product graininess.

Surrogate responses are common, but not without risks. In particular, we may find that the surrogate response turns out not to be a good predictor of the primary response.

Cardiac arrhythmias

Acute cardiac arrhythmias can cause death. Encainide and flecanide acetate are two drugs that were known to suppress acute cardiac arrhythmias and stabilize the heartbeat. Chronic arrhythmias are also associated with sudden death, so perhaps these drugs could also work for nonacute cases. The Cardiac Arrhythmia Suppression Trial (CAST) tested these two drugs and a placebo (CAST Investigators 1989). The real response of interest is survival, but regularity of the heartbeat was used as a surrogate response. Both of these drugs were shown to regularize the heartbeat better than the placebo did. Unfortunately, the real response of interest (survival) indicated that the regularized pulse was too often 0. These drugs did improve the surrogate response, but they were actually worse than placebo for the primary response of survival.

By the way, the investigators were originally criticized for including a placebo in this trial. After all, the drugs were *known* to work. It was only the placebo that allowed them to discover that these drugs should not be used for chronic arrhythmias.

In addition to responses that relate directly to the questions of interest, some experiments collect *predictive* responses. We use predictive responses to model the primary response. The modeling is done for two reasons. First, such modeling can be used to increase the precision of the experiment and the comparisons of interest. In this case, we call the predictive responses *covariates* (see Chapter 17). Second, the predictive responses may help us understand the mechanism by which the treatment is affecting the primary response. Note, however, that since we observed the predictive responses rather than setting them experimentally, the mechanistic models built using predictive responses are observational.

A final class of responses is *audit* responses. We use audit responses to ensure that treatments were applied as intended and to check that environmental conditions have not changed. Thus in a study looking at nitrogen fertilizers, we might measure soil nitrogen as a check on proper treatment application, and we might monitor soil moisture to check on the uniformity of our irrigation system.

Example 1.2

Predictive
responses

Audit responses

